

Unit II: Laws of Motion and Forces

Newton's laws, friction, tension, normal force, free body diagrams, and circular motion

Learning Goals

- Understand force as an interaction that can change the motion or shape of a body.
- Apply Newton's three laws of motion to everyday and numerical problems.
- Identify common forces such as weight, normal force, friction, and tension.
- Draw and use free body diagrams to solve force problems systematically.
- Analyze motion on horizontal surfaces, inclined planes, and connected bodies.
- Understand centripetal acceleration and centripetal force in circular motion.
- Avoid common misconceptions about action-reaction pairs, friction, and circular motion.

1. Force and Motion

A force is a push or pull acting on a body because of its interaction with another body. Force is a vector quantity, so it has both magnitude and direction. The SI unit of force is the newton, written as N.

$$1 \text{ newton} = 1 \text{ kg m/s}^2$$

Forces do not always produce motion. A book resting on a table has forces acting on it, but the forces balance, so the book remains at rest. Motion changes only when there is a nonzero net force.

- A balanced force system has zero net force and no acceleration.
- An unbalanced force system has nonzero net force and produces acceleration.
- The net force is the vector sum of all forces acting on the object.

$$F_{\text{net}} = \text{sum of all forces}$$

2. Newton's First Law

Newton's first law states that an object remains at rest or continues moving with uniform velocity in a straight line unless acted upon by a nonzero external net force. This law is also called the law of inertia.

Inertia

Inertia is the tendency of a body to resist changes in its state of motion. Mass is a measure of inertia. A larger mass is harder to start, stop, or change in direction.

- A passenger lurches forward when a bus suddenly stops because the passenger's body tends to continue moving forward.
- A coin falls into a glass when a card beneath it is flicked away quickly because the coin tends to remain at rest.
- A spacecraft in deep space can continue moving without fuel if no significant external force acts on it.

The first law also introduces the idea of an inertial frame of reference. An inertial frame is one in which Newton's laws hold without needing fictitious forces.

3. Newton's Second Law

Newton's second law gives the quantitative relation between force, mass, and acceleration. The acceleration of an object is directly proportional to the net external force and inversely proportional to its mass.

$$F_{\text{net}} = m a$$

- Acceleration is in the direction of the net force.
- If the same force acts on a larger mass, the acceleration is smaller.
- If the net force is doubled while mass is constant, acceleration is doubled.
- If $F_{\text{net}} = 0$, then $a = 0$; the object is either at rest or moving with constant velocity.

Component Form

In two-dimensional problems, Newton's second law is applied separately along perpendicular axes. Usually the axes are chosen to make the problem simpler.

$$\sum F_x = m a_x \qquad \sum F_y = m a_y$$

For an inclined plane, it is often easiest to choose one axis parallel to the plane and the other perpendicular to the plane. This reduces the number of forces that must be resolved into components.

4. Newton's Third Law

Newton's third law states that for every action force, there is an equal and opposite reaction force. These two forces act on different bodies, so they do not cancel each other on the same free body diagram.

$$F_{AB} = -F_{BA}$$

- If a hand pushes a wall, the wall pushes the hand back with equal magnitude and opposite direction.
- Earth pulls a falling object downward; the object pulls Earth upward with equal gravitational force.
- A rocket pushes gases backward; the gases push the rocket forward.

Important Point

Action-reaction forces never act on the same object. This is why they do not cancel when analyzing the motion of one body. Only forces acting on the chosen body are included in that body's free body diagram.

5. Common Types of Forces

Weight

Weight is the gravitational force exerted by Earth or another planet on a body. Near Earth's surface, weight acts vertically downward and has magnitude mg .

$$W = mg \quad g \text{ approximately } 9.8 \text{ m/s}^2$$

- Mass is the amount of matter and is measured in kilograms.
- Weight is a force and is measured in newtons.
- Mass remains the same from place to place, but weight changes if g changes.

Normal Force

The normal force is the contact force exerted by a surface perpendicular to that surface. It prevents objects from passing through each other. Normal force is not always equal to weight.

- On a horizontal surface with no vertical acceleration, $N = mg$.
- On an inclined plane, $N = mg \cos(\theta)$ if no other perpendicular forces act.
- In an elevator accelerating upward, the normal force is greater than mg .
- In an elevator accelerating downward, the normal force is less than mg .

Tension

Tension is the pulling force transmitted through a string, rope, or cable. In ideal problems, the string is massless and inextensible, and the pulley is frictionless. Under these assumptions, tension is the same throughout one continuous string.

- Tension always pulls; it does not push.
- Tension acts along the direction of the string.
- Connected-body problems often use the same acceleration for bodies joined by a taut string.

Friction

Friction is a contact force that opposes relative motion or the tendency of relative motion between surfaces. It acts parallel to the surface of contact.

$$\text{Static friction: } f_s \leq \mu_s N \quad \text{Kinetic friction: } f_k = \mu_k N$$

- Static friction acts when surfaces are not sliding relative to each other.
- Kinetic friction acts when surfaces are sliding.
- Static friction adjusts up to a maximum value; it is not always equal to $\mu_s N$.
- Usually μ_s is greater than μ_k , so starting motion is harder than maintaining sliding motion.
- Friction opposes relative motion or impending relative motion, not necessarily the object's velocity relative to the ground.

6. Free Body Diagrams

A free body diagram is a diagram showing all external forces acting on a selected object. It is one of the most important tools for solving mechanics problems.

Steps for Drawing a Free Body Diagram

- Choose the object or system to analyze.
- Represent the object as a point or simple shape.
- Draw only external forces acting on that object.
- Label every force clearly: weight, normal force, tension, friction, applied force, and so on.

- Choose convenient coordinate axes.
- Resolve angled forces into components when needed.
- Apply $\sum F_x = ma_x$ and $\sum F_y = ma_y$.

Forces Usually Included

- Weight acts downward through the center of mass.
- Normal force acts perpendicular to the contact surface.
- Friction acts parallel to the surface and opposes sliding or tendency of sliding.
- Tension acts along a string and pulls away from the object.
- Applied force acts in the direction in which the push or pull is applied.

Forces Not Included

- Do not include forces exerted by the object on other bodies.
- Do not include action-reaction force pairs on the same diagram unless both act on the chosen body, which they usually do not.
- Do not invent a force in the direction of motion. Motion can continue without a forward force if acceleration is zero.

7. Solving Force Problems

Most force problems follow a common pattern: draw the free body diagram, choose axes, write Newton's second law in each direction, substitute known values, and solve unknowns.

Horizontal Surface

For a block on a horizontal surface pulled by a horizontal force, the vertical forces often balance if there is no vertical acceleration. The horizontal equation determines acceleration.

$$\text{Vertical: } N - mg = 0$$

$$\text{Horizontal: } F - f = ma$$

Inclined Plane

On an inclined plane, weight is resolved into two components. The component parallel to the plane is $mg \sin(\theta)$, and the component perpendicular to the plane is $mg \cos(\theta)$.

$$\text{Parallel component: } mg \sin(\theta)$$

$$\text{Perpendicular component: } mg \cos(\theta)$$

$$\text{Normal force on simple incline: } N = mg \cos(\theta)$$

- If the plane is frictionless, acceleration down the plane is $g \sin(\theta)$.
- If friction is present, friction acts opposite the motion or tendency of motion.

Connected Bodies

For bodies connected by a light string, each body gets its own free body diagram. If the string remains taut, connected bodies often share the same acceleration magnitude.

- Write separate equations for each mass.
- Use the same tension for a single ideal string.
- Use signs consistently according to the chosen motion direction.

8. Circular Motion

Circular motion occurs when an object moves along a circular path. Even if speed is constant, velocity changes continuously because direction changes. Therefore, an object in circular motion has acceleration.

Uniform Circular Motion

Uniform circular motion means the object moves in a circle with constant speed. The acceleration is directed toward the center of the circle and is called centripetal acceleration.

$$\text{Centripetal acceleration: } a_c = v^2 / r$$

$$\text{Centripetal force: } F_c = m v^2 / r$$

- Centripetal means center-seeking.
- Centripetal force is not a new separate force; it is the net inward force required for circular motion.
- The inward force may be provided by tension, friction, gravity, normal force, or a combination of forces.

Period and Frequency

The period is the time taken for one complete revolution. Frequency is the number of revolutions per second.

$$v = 2 \pi r / T \qquad f = 1 / T \qquad v = 2 \pi r f$$

Examples of Centripetal Force

- A stone tied to a string: tension provides centripetal force.
- A car turning on a flat road: friction provides centripetal force.
- A satellite orbiting Earth: gravity provides centripetal force.
- A roller coaster at the bottom of a loop: normal force and weight combine to produce net inward force.

9. Non-Uniform Circular Motion

In non-uniform circular motion, speed changes as the object moves. The acceleration has two components: centripetal acceleration toward the center and tangential acceleration along the tangent.

$$\text{Radial acceleration: } a_r = v^2 / r \qquad \text{Tangential acceleration: } a_t = dv/dt$$

- Radial acceleration changes the direction of velocity.
- Tangential acceleration changes the speed.
- The total acceleration is the vector combination of radial and tangential acceleration.

10. Worked Examples

Example 1: Net Force

A 5 kg box is pulled with a horizontal force of 20 N on a frictionless floor. Using $F_{\text{net}} = ma$, acceleration $a = 20/5 = 4 \text{ m/s}^2$.

Example 2: Friction

A 10 kg block slides on a surface with $\mu_k = 0.2$. The normal force is $N = mg = 10 \times 9.8 = 98 \text{ N}$. Kinetic friction is $f_k = \mu_k N = 0.2 \times 98 = 19.6 \text{ N}$.

Example 3: Inclined Plane

For a block on a frictionless incline of angle θ , the component of weight down the plane is $mg \sin(\theta)$. Newton's second law gives $mg \sin(\theta) = ma$, so $a = g \sin(\theta)$.

Example 4: Circular Motion

A 2 kg object moves in a circle of radius 1 m with speed 3 m/s. Centripetal force $F_c = mv^2/r = 2 \times 9 / 1 = 18 \text{ N}$ toward the center.

11. Common Misconceptions

- A moving object does not need a force in the direction of motion to keep moving at constant velocity.
- Action and reaction forces do not cancel because they act on different bodies.
- Normal force is not always equal to weight.
- Static friction is not always equal to $\mu_s N$; it adjusts up to its maximum value.
- Centripetal force is not an extra mysterious force; it is the net inward force.
- In uniform circular motion, speed is constant but velocity is not constant because direction changes.
- At the top of a vertical circle, both weight and possibly tension/normal force may point toward the center.

Quick Revision Sheet

- Force is a vector measured in newtons.
- Newton's first law describes inertia.
- Newton's second law is $F_{\text{net}} = ma$.
- Newton's third law says action and reaction are equal, opposite, and act on different bodies.
- Weight is mg and acts downward.
- Normal force is perpendicular to the surface.
- Tension acts along a string and always pulls.
- Friction acts parallel to the surface and opposes relative motion or tendency of motion.
- Static friction satisfies $f_s \leq \mu_s N$; kinetic friction is $f_k = \mu_k N$.
- A free body diagram contains only external forces on the selected object.
- On an incline, weight components are $mg \sin(\theta)$ and $mg \cos(\theta)$.
- Uniform circular motion has centripetal acceleration v^2/r toward the center.
- Centripetal force is the net inward force required for circular motion.

Practice Questions

- State Newton's three laws of motion with one example for each.
- Why does a passenger move forward when a bus suddenly stops?
- A 4 kg object experiences a net force of 12 N. Find its acceleration.
- Differentiate between mass and weight.
- Explain why normal force is not always equal to mg .
- A 20 kg box rests on a horizontal floor. Find its weight and normal force.
- A block has $N = 50 \text{ N}$ and $\mu_k = 0.3$. Find kinetic friction.
- What is the maximum static friction if $\mu_s = 0.5$ and $N = 80 \text{ N}$?
- Draw the free body diagram of a block pulled by a string on a rough horizontal surface.
- Resolve the weight of a block on an incline into parallel and perpendicular components.
- Why do action-reaction pairs not cancel each other?
- A 2 kg body moves in a circle of radius 0.5 m with speed 4 m/s. Find centripetal force.
- What provides centripetal force for a car turning on a flat road?
- Explain why uniform circular motion is accelerated motion.
- Differentiate between radial acceleration and tangential acceleration.